

Graphs II

Previous Lecture

- *Graphs, directed and undirected*
- *Adjacency, incidence, endpoints*
- *Degree of vertices, isolated and pendant vertices*
- *Walks, trails, and paths*
- *Bacon numbers, Erdős numbers:*
 - *Length of **shortest path** connecting someone to Kevin Bacon in the Hollywood graph (or mathematician Paul Erdős in the Academic co-author graph)*
 - *My Erdos number is 4 (but my Bacon number is ∞ ...)*
 - *Andrei's Erdos number is 3*
 - *Pavol Hell (professor emeritus of CS here) has Erdos number 1*

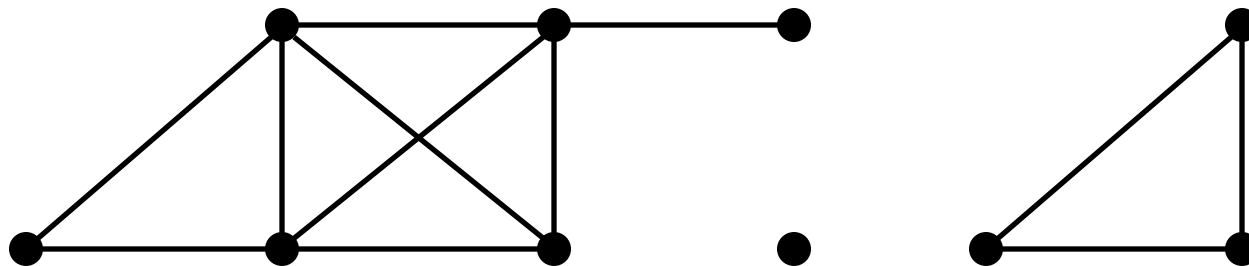
Connected Components

● Recall:

- A **walk** is a sequence of alternating vertices and edges
- A **path** is a walk with no repeated vertices

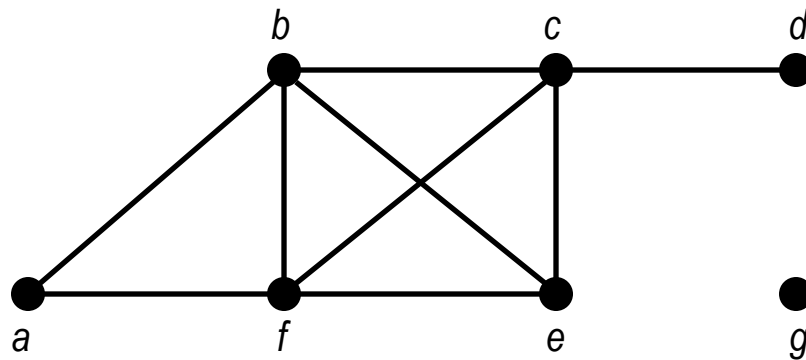
● An undirected graph $G = (V, E)$ is said to be **connected** if there is a path between any two distinct vertices of G

● Let $G = (V, E)$ be an undirected graph. A maximal (with respect to inclusion) set of vertices W such that there is a path between any two distinct vertices from W is called a **connected component** of G

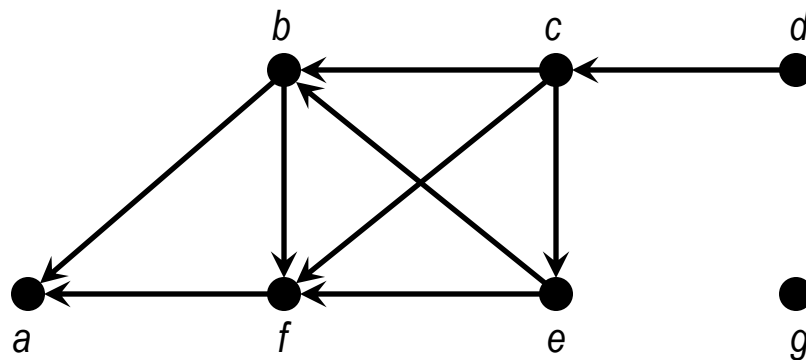


Representation of Graphs: Lists of Vertices and Edges

● *Sets of vertices and edges*



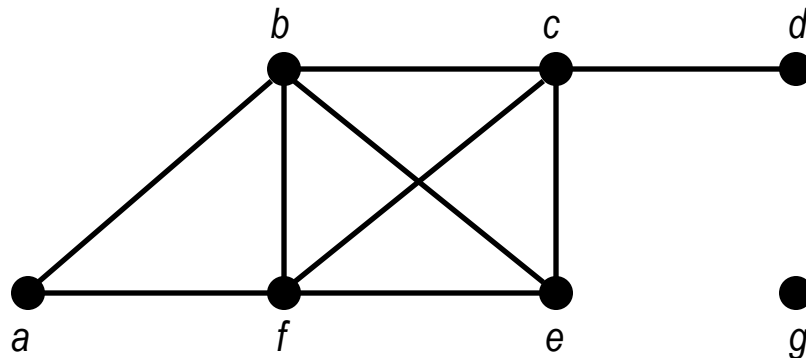
$\{a, b, c, d, e, f, e, g\}$
 $\{ \{a,b\}, \{b,c\}, \{a,f\}, \{b,f\}, \{f,c\},$
 $\{b,e\}, \{c,e\}, \{c,d\} \}$



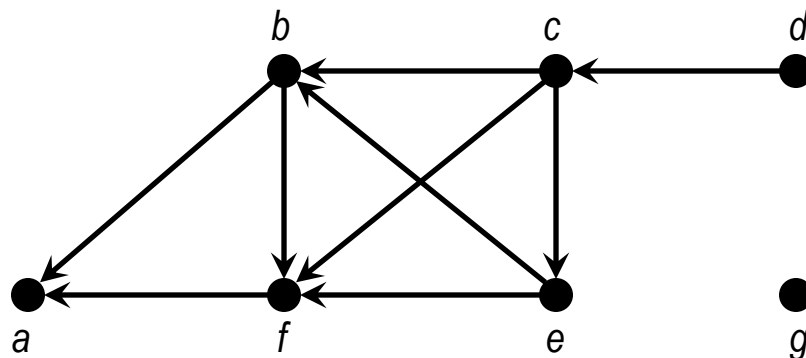
$\{a, b, c, d, e, f, e, g\}$
 $\{ (b,a), (c,b), (f,a), (b,f), (c,f),$
 $(e,b), (c,e), (d,c) \}$

Representation of Graphs: Adjacency Lists

● *Adjacency list is a table containing for each vertex a set of vertices adjacent to it.*



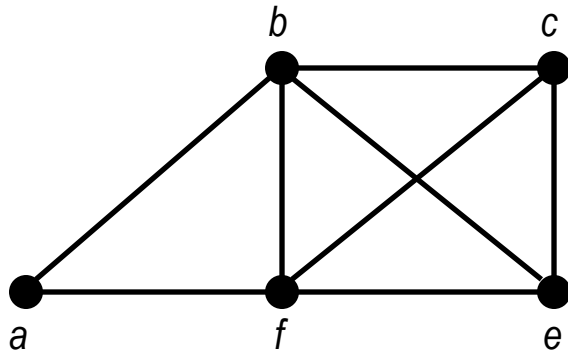
vert.	adjacent	vert.	adjacent
a		e	
b		f	
c		g	
d			



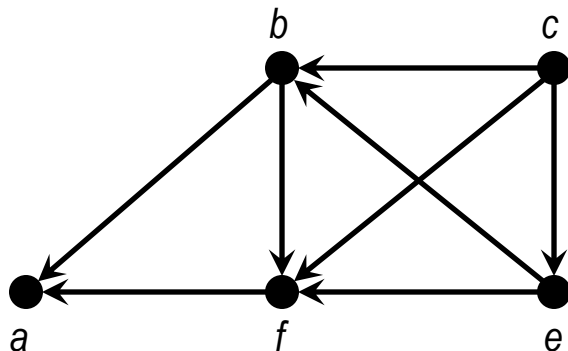
vert.	adjacent	vert.	adjacent
a		e	
b		f	
c		g	
d			

Representation of Graphs: Adjacency Matrix

- The adjacency matrix of a graph is a 0-1-matrix whose rows and columns are labeled by the vertices of the graph. The entry in i th row and j th column is 1 if there is an edge from vertex i to vertex j . Otherwise it is 0.



$$\begin{matrix} & a & b & c & e & f \\ \begin{matrix} a \\ b \\ c \\ e \\ f \end{matrix} & \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \end{matrix}$$



$$\begin{matrix} & a & b & c & e & f \\ \begin{matrix} a \\ b \\ c \\ e \\ f \end{matrix} & \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix} \end{matrix}$$

Example

● Draw a graph with the adjacency matrices

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Example

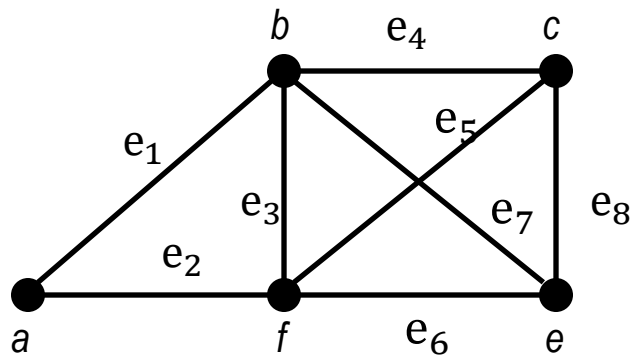
● *The adjacency matrix of a multigraph shows how many edges connect every pair of vertices.*

● *Draw a multigraph with the adjacency matrix*

$$\begin{pmatrix} 0 & 3 & 0 & 2 & 1 \\ 3 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 3 \\ 2 & 0 & 2 & 3 & 0 \end{pmatrix}$$

Representation of Graphs: Incidence Matrix

- The incidence matrix of a graph $G = (V, E)$ is a 0-1-matrix whose rows are labeled by vertices of G and columns are labeled by edges of G . The entry in i th row and j th column is 1 if the vertex i is incident to the edge j .



$$\begin{pmatrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ a & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ b & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ c & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ e & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ f & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Example

● Draw a graph with the incidence matrix

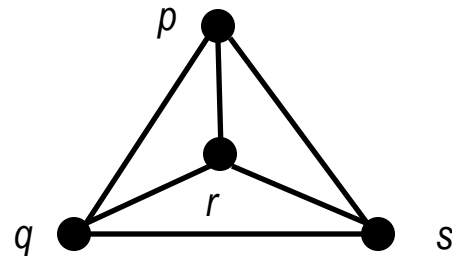
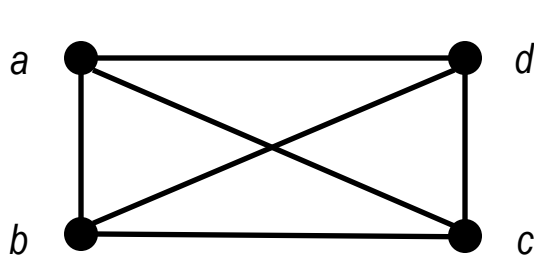
$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

Comparing representations

- *Which method is better?*
- *It depends on a graph **and the problem we are going to solve**.*
- *If a graph is sparse (e.g. constant number of edges per vertex), then the adjacency list and incidence matrix are more concise.*
- *If a graph is dense, that is, contains many edges, then adjacency matrix is more concise.*
- *Some algorithms may require particular type of representation.*
- *Problem (CONNECTIVITY):
 Compute the connected components of a graph*
- *On an adjacency matrix or incidence matrix, CONNECTIVITY can be solved efficiently by matrix multiplication*

Isomorphism of Graphs

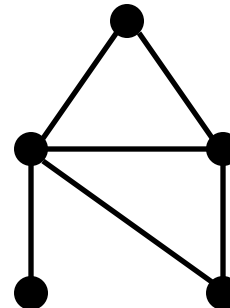
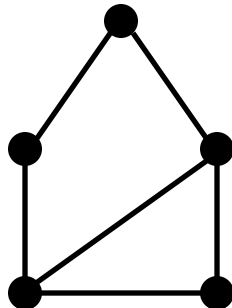
- We often need to know whether it is possible to draw two graphs in the same way
- Simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if there is a bijection f from V_1 to V_2 such that vertices a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2
- Such a function f is called an **isomorphism**.



x	a	b	c	d
$f(x)$	r	q	s	p

Graph Invariants

- In general, it is **very hard** to decide whether two graphs are isomorphic or not.
 - At least as hard as factoring integers, it seems
- We can however often prove that two graphs are **not** isomorphic by finding a property that one graph has which the other does
- Example: Are these graphs isomorphic?

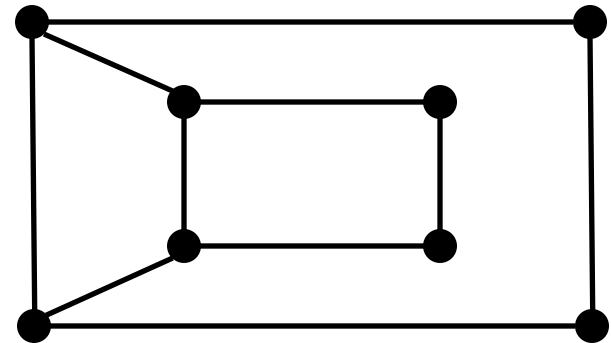
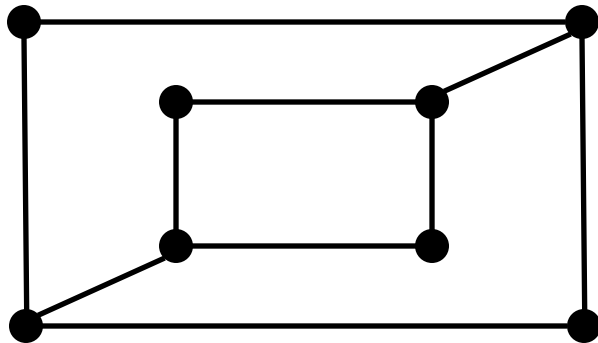


- A property preserved by isomorphism of graphs is called a **graph invariant**.

More invariants

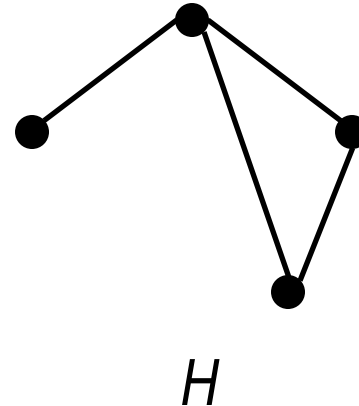
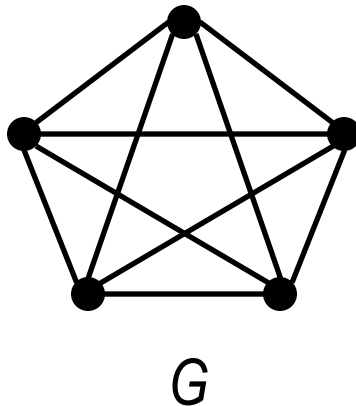
- *Graph invariants include:*
 - number of vertices & edges*
 - number of connected components*
 - degrees of vertices*
 - the length of paths and cycles*

Example: Are these graphs isomorphic?



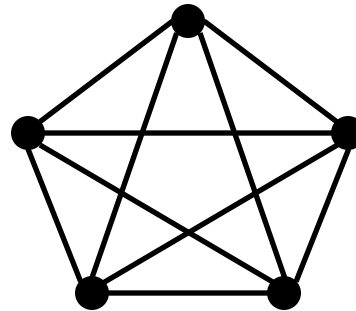
Subgraphs

- Sometimes we need only a part of a big graph.
- A **subgraph** of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$.



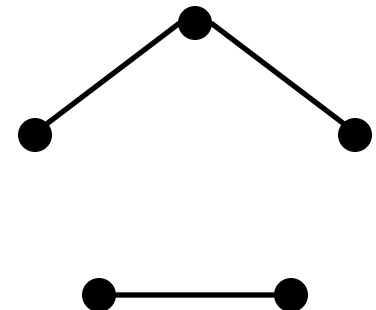
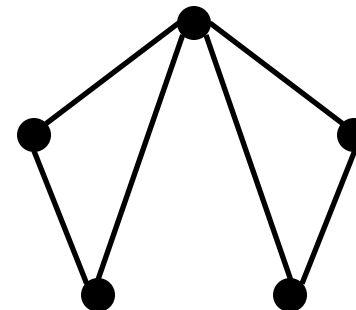
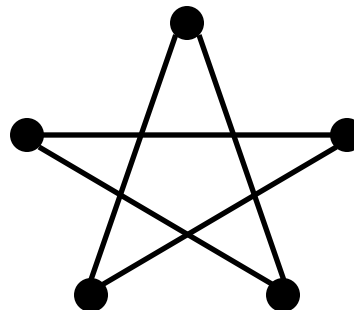
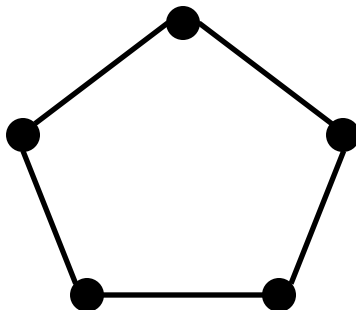
Subgraphs (cntd)

- If $H = (W, F)$ is a subgraph of $G = (V, E)$ and $W = V$ then H is called a **spanning subgraph**



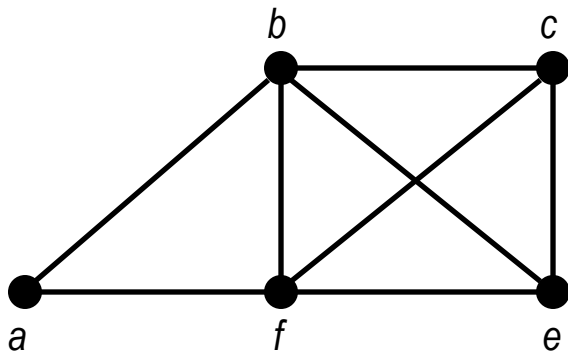
G

Spanning subgraphs:

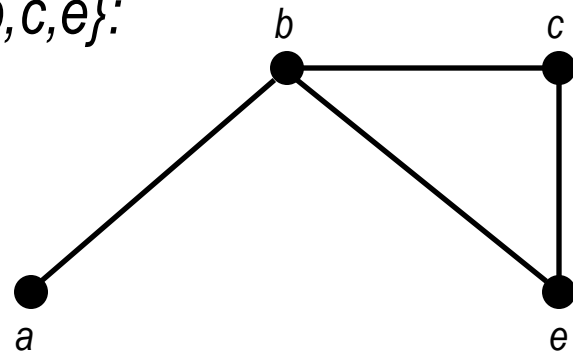


Subgraphs (cntd)

- Let $G = (V, E)$ be a graph and $W \subseteq V$ a nonempty set of vertices. The **subgraph** of G **induced** by W is the subgraph whose vertex set is W and which contains all edges (from G) whose endpoints are in W .
- A subgraph H of a graph $G = (V, E)$ is called an induced subgraph if there is $W \subseteq V$ such that H is the subgraph induced by W .

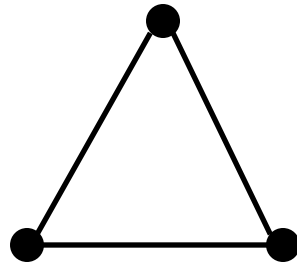
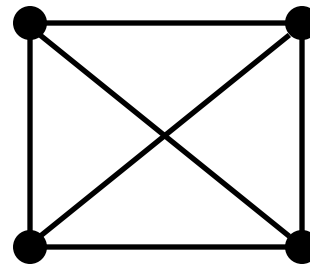
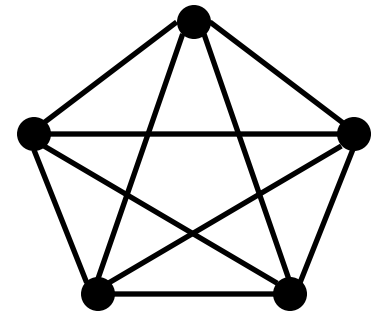


$W = \{a, b, c, e\}$:



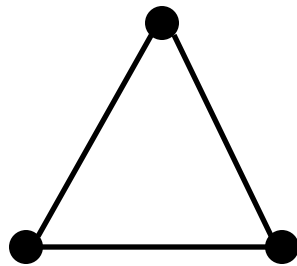
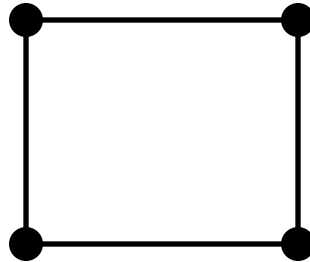
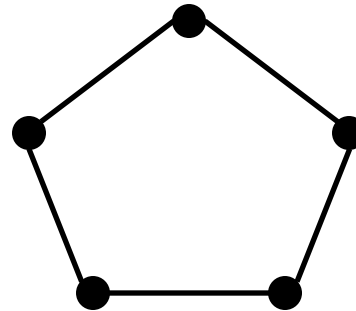
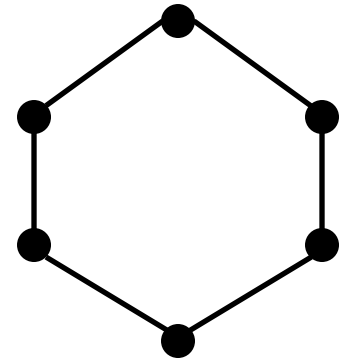
Some Special Graphs

- *We're often interested in whether a graph (or subgraph) is isomorphic to a graph with a particular structure*
- *This tells us a lot about the properties and structure of the graph*
- **Complete Graphs.** *The complete graph K_n on n vertices is the simple graph that contains exactly one edge between each pair of distinct vertices*

 K_1  K_2  K_3  K_4  K_5

Some Special Graphs (cntd)

- **Cycles.** The cycle C_n for $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.

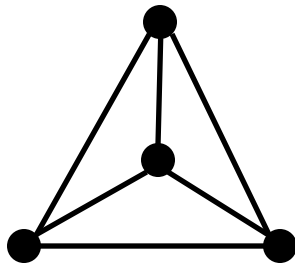
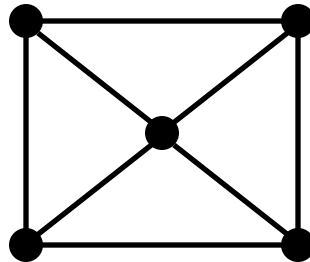
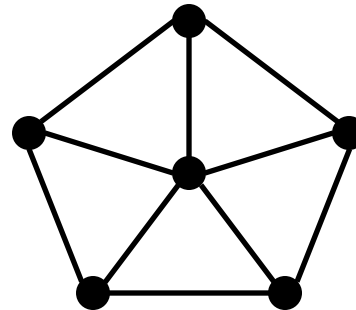
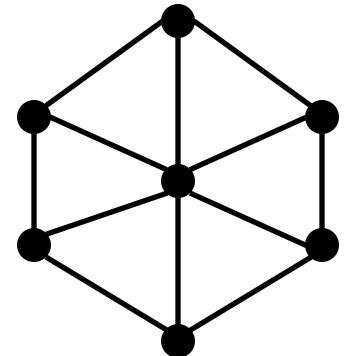
 C_3  C_4  C_5  C_6

- **Theorem:**

A graph has a loop of length k if and only if it has a subgraph isomorphic to C_k

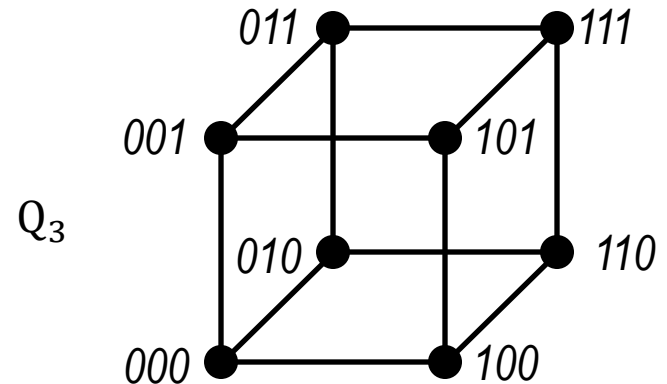
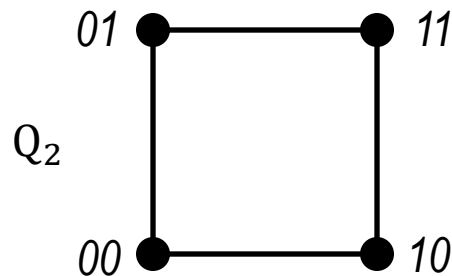
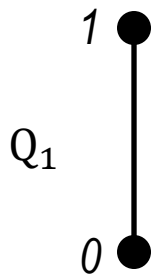
Some Special Graphs (cntd)

- *Wheels.* The wheel, w_n , is obtained from the cycle c_n when we add an additional vertex to it and connect this vertex to each of the n vertices in c_n by new edges

 W_3  W_4  W_5  W_6

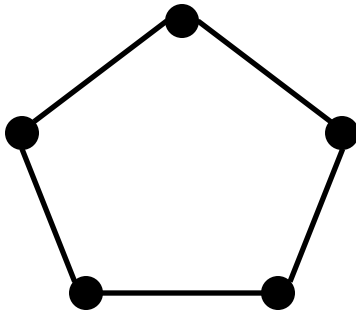
Some Special Graphs (cntd)

- *Hypercubes. The n -dimensional hypercube, or n -cube, denoted by Q_n is the graph that has vertices representing the bit strings of length n . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.*

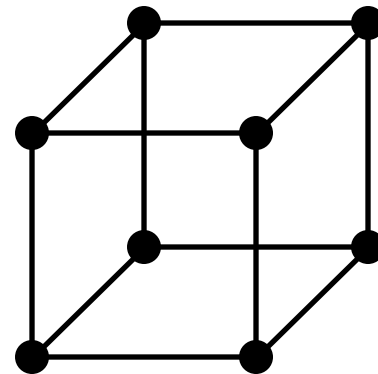


Regular Graphs

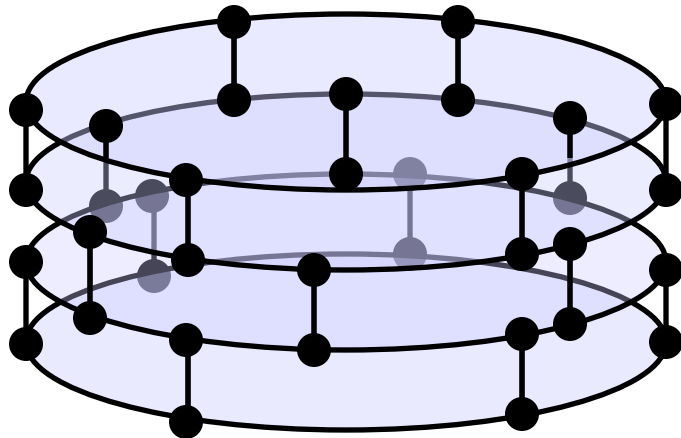
● A graph is called *k-regular* if all its vertices have degree k .



2-regular graphs are cycles



The k -cube is a k -regular graph



A brick graph, another example of a 3-regular graph

Practice

Exercises from the Book:

7th edition: 26, 53, 55 (page 666 – 667), 1, 3, 5, 7, 9c, 11, 39 (page 675 – 676)

8th edition: 26, 55, 57, (page 699 – 700), 1, 3, 5, 7, 9c, 11, 43 (page 710 – 712)